Optimizing Kidney Stone Detection with Vision Transformers and CNNs: A Transfer Learning Strategy

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Abstract— Kidney stones are a common and impactful urological condition, with timely and accurate diagnosis being critical to prevent complications such as renal damage and avoid invasive treatments. This study explores the use of advanced deep learning models, including Vision Transformers (ViT), Swin Transformer and Convolutional Neural Networks (CNNs)—specifically XResNet-50, VGG-16, MobileNetV2 and AlexNet—to enhance kidney stone detection from CT scan images. Leveraging a robust dataset of annotated CT scans in JPG format, the approach employs transfer learning which leverages features learnt from the ImageNet dataset to optimize model performance, focusing on classifying cases as either "Kidney Stone Present" or "Kidney Stone Absent". To further improve diagnostic reliability, the Grad-CAM (Gradient-weighted Class Activation Mapping) identifies and localizes kidney stones within CT images, enhancing understanding of the model's decision-making and supporting interpretability. Addressing challenges related to dataset imbalance, model interpretability, and generalizability across healthcare settings. By refining these models for kidney stone detection, the main objective of this work is to build a reliable diagnostic tool that aids in clinical judgement and treatment planning, ultimately enhancing patient outcomes and encouraging broader clinical adoption.

Keywords— CT - Computed Tomography Scan, Kidney Stone, Vision Transformers, Convolutional Neural Network, Transfer Learning, GRAD - CAM.

I. INTRODUCTION

Kidney stones, or renal calculi, are crystalline minerals and salts deposits that form within the kidneys, affecting millions of people worldwide. These stones can obstruct urinary pathways, leading to symptoms such as severe pain, hematuria, and, in more severe cases, infections and renal failure. Detecting kidney stones early is critical because timely treatment and care can help minimize complications and reduce the need for invasive procedures such as lithotripsy or surgical removal. Traditional diagnostic methods, including ultrasound and CT imaging, are effective but often require specialized interpretation, making accurate diagnosis time-consuming and resource-intensive.

The kidneys(Fig. 1) are bean-shaped organs located on each side of the spine, positioned behind the abdomen and just below the rib cage. Their primary function is to filter blood, maintaining the balance of bodily fluids by removing waste materials [1]. When kidney stones—hard deposits of minerals and salts—grow to a size of 3 millimeters or larger, they can obstruct the ureter, leading to severe pain, typically in the lower back, which may radiate to the groin. Kidney dysfunction may lead to serious health issues, underscoring the importance of early detection of kidney stones [2]. Accurate identification of urinary calculi is essential to improve treatment outcomes and increase the success rate of surgical interventions. Common diagnostic tools for kidney stone detection include computed tomography (CT), ultrasound, and blood tests. In this study, CT scans are utilized as shown in Fig. 2, as they offer high-resolution, cross-sectional images that provide detailed information about the location, and density of stones, making them especially effective for accurate detection and assessment as shown in Fig. 3. Accurate identification of urinary calculi is essential to improve treatment outcomes and increase the success rate of surgical interventions.

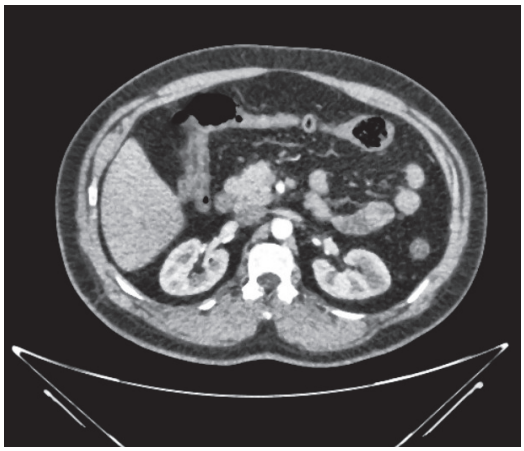

Fig. 1. Normal kidney CT image

By leveraging state-of-the-art deep learning algorithms, we present a prediction model aimed at enhancing the accuracy for stone detection, with a secondary goal to identify and localize kidney stones within CT images using GRAD-CAM. This model supports healthcare professionals in

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making faster and more precise decisions, ultimately improving patient outcomes and reducing the risks associated with delayed diagnosis.

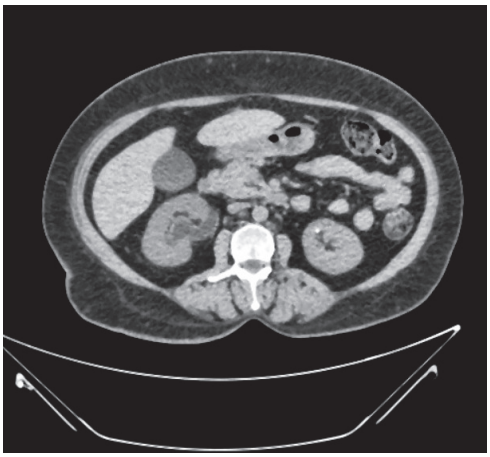

Fig. 2. Kidney stone CT image

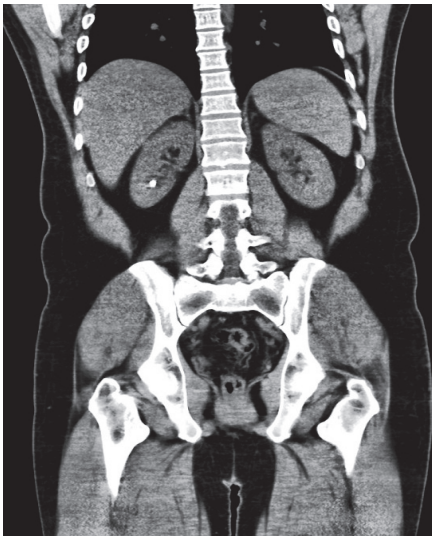

Fig. 3. Vertical cross-sectional view of kidney stone CT image

II. Related Works

A multitude of research have yielded varied results related to kidney stone detection. A sample study involving 40 patients was conducted to evaluate the effectiveness of kidney stone detection using CT scan images. The methodology included preprocessing with a median filter to reduce noise, segmentation using the k-means clustering algorithm to isolate relevant regions, followed by the detection and classification of kidney stones. Data was collected from a hospital setting, ensuring clinical relevance. The results of this study demonstrated a promising diagnostic accuracy of 92.5%, highlighting the potential of this approach[3].

Another study focuses on the binary categorization of kidney CT images utilizing a pre-trained VGG16 model, followed by Fuzzy C-Means clustering to identify areas of interest. For single image predictions, a median filter is used to remove noise from CT images, while Adaptive Histogram Equalization (AHE) is used to enhance image contrast. The approach achieved an accuracy of 88% using Fuzzy C-Means clustering; however, the approach has several limitations, such as sensitivity to noise and difficulty in precisely segmenting irregularly shaped kidney stones. These limitations underscore the requirement for more comprehensive segmentation techniques to improve detection accuracy[4].

In a different study, researchers explored kidney stone detection through a deep learning method using a CNN model utilizing the Inception V3 architecture, trained using small dataset of 5,200 images. The model achieved an accuracy of only 61.6%, revealing a significant drawback: the insufficient number of images likely hindered the model's capacity to learn complex patterns and generalize effectively. This finding highlights the necessity for larger, high-quality datasets to improve accuracy and reliability in deep learning-based medical imaging tasks[5].

To detect kidney stones, a XResNet50 CNN model with CT scan images in the axial, coronal, and sagittal planes was used with 2,709 stone-present and 250 stone-free pictures. The model achieved 78%, 63%, and 85% accuracy rates in the axial, coronal, and sagittal planes, respectively, for stones measuring 0-1 cm. Accuracy increased in sagittal views for larger stones, reaching 93% for stones larger than 2cm[6].

Another study uses traditional machine learning approaches (Random Forest, XGBoost, and SVM) and deep learning approaches (AlexNet and Inception v3) to classify kidney stones in endoscopic pictures. Shallow models use handmade color and texture attributes, whereas deep learning models use knowledge transfer and data augmentation to manage a small dataset. Inception v3 had the strongest performance, with precision, recall, and F1-scores of up to 97%. Shallow approaches, such as XGBoost, achieved 96% accuracy but suffered in complex imaging circumstances. ResNet-101 struggled to detect finer textures in certain stone types, such as struvite and brushite, and performed less consistently in real-time, variable-quality video[7].

III. Proposed Methodology

A. Dataset Description

The particular dataset used in this study consists of CT scan images sourced from both public platforms such as Kaggle and Github in addition to private sources, specifically Unity Lifeline Hospital. The dataset is separated into two classes: Normal, containing 5,915 images, and Stone, with 3,495 images. To prepare the data for model training, a 70-30 train-test split was applied.

B. Data Augmentation

To mitigate overfitting and enhance model generalizability, data augmentation techniques were employed, including rescaling and altering image orientation, effectively increasing the diversity and volume of training images[8].

Data Augmentation as shown in Fig. 4, is the process of augmenting the available data by transforming certain aspects of the image data so that the neural network gets to see a lot of images which reduces the chance of overfitting in training. Augmentation techniques like rotation, zooming, and flip- top to bottom and left to right, were used to mitigate data imbalance throughout the training and verification of our model.

C. Data Preprocessing

For effective image preprocessing, techniques such as Median Blur and Contrast Limited Adaptive Histogram Equalization (CLAHE) were applied exclusively to the

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training dataset using the Torch FastAI library. The raw images, initially received from lab technicians, underwent resizing and standardization transformations to ensure consistency with the existing dataset. Identifiable patient information has been excluded to safeguard privacy and maintain anonymity.

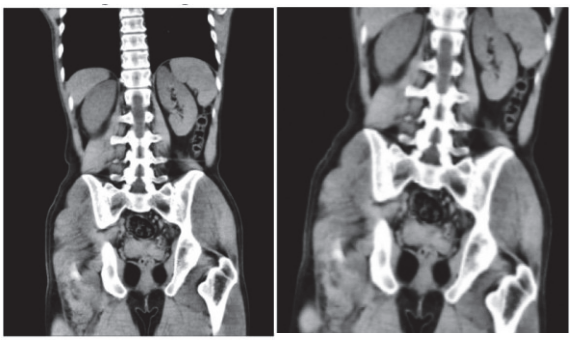

Fig. 4. (a): Original Image (b): Augmented Image

Median Blur was used to reduce speckle noise which is common in CT scans by replacing each pixel's value with the median value of neighboring pixels, effectively smoothing out minor artifacts without blurring critical edges. CLAHE was applied to enhance image contrast by adjusting local histogram distributions, making subtle features more distinguishable and supporting the accurate detection of kidney stones.

This preprocessing process ensured the dataset's diagnostic relevance and usability, while upholding patient confidentiality.

D. Approach

The proposed system intends to optimize Kidney stone detection through Convolution Neural Networks (CNN) trained on the diverse dataset comparing on different models such as VGG16, Alexnet, MobileNet, Resnet50, SwinTransformer, Resnet50+ViT hybrid model.

The transfer learning approach is particularly beneficial for smaller datasets, as it leverages pre-trained models that are often initially trained on large and diverse datasets like ImageNet to extract useful low- and mid-level features. Fine-tuning these models enables huge savings in time and computing resources, since the model has already learned general representations that can be adapted to the particular task at hand. This approach enhances performance, especially when data is limited, as the model is able to generalize better and requires fewer labeled examples to attain high accuracy. By starting with pre-trained weights, a more effective model can be built more quickly, making transfer learning a powerful strategy.

The Fig 5 illustrates a kidney stone detection algorithm using CT images. It begins with data collection and preprocessing, including augmentation, to reduce overfitting while training. A pre-trained ResNet50 model extracts features, followed by Vision Transformers for global attention. Global attention enables the model to attend to relevant regions across the entire spatial domain of an image, capturing contextual dependencies critical in medical imaging. A hybrid architecture leveraging ResNet50 and Vision Transformers (ViTs) combines their respective strengths. ResNet50's convolutional layers effectively extract localized, hierarchical feature representations, while ViTs use self-attention mechanisms to model long-range dependencies. This synergy allows the model to learn both fine-grained spatial features and global contextual relationships, enhancing accuracy and robustness in complex tasks like kidney stone detection, where both localized patterns and spatial context are essential for precise classification. The features obtained through Resnet-50 are fed into VIT for classification and for predicting whether a kidney stone is "Present" or "Absent". The model's performance is compared to other models for evaluation. To enhance interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) generates heatmaps, highlighting regions the model considers, helping clinicians validate detected stones' presence and location, increasing transparency in predictions.

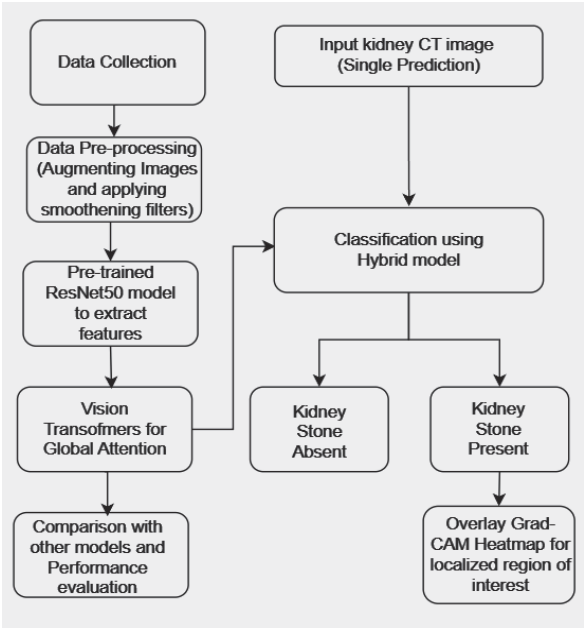

Fig. 5. Architectural diagram of Resnet+ViT

IV. Experimental Analysis And Results

A. Experimental Setup

The experiment was implemented on Kaggle's notebook environment using Python 3.12.0, with computations executed on a T4 GPU runtime. The data was pre-processed and analyzed using NumPy, Matplotlib, and Pandas libraries.

The model evaluation procedures were conducted with reference to the parameters and values outlined by Table I.

The hidden layer employs ReLU (Rectified Linear Unit) which is an activation function that introduces non-linearity, enabling the model to capture more complex patterns. The output layer uses the Softmax activation function, commonly applied for multi-class classification tasks, providing a probability distribution over possible classes. The loss function is Categorical Cross entropy, which measures the performance of a classification model where the output is a one-hot encoded label.

TABLE I. PARAMETERS AND VALUES OF NEURAL NETWORK

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Parameters	Values
Hidden Layer activation function	Relu
Output Layer activation function	Softmax
Loss Function	Categorical-Cross entropy
Batch Size	32
Epochs for Resnet50	40
Epochs for Resnet+Vit	10
Epochs for XResnet	20
Epochs for VGG-16	20
Epochs for Alexnet	20
Epochs for MobileNet	20
Epochs for SwinTransformer	20
Optimizer	ADAM

V. OBSERVATIONS

This paper presents a comprehensive evaluation of several approaches for kidney stones detection on CT scans using XResNet, AlexNet, VGG, Mobilenet, Swin Transformer, and a hybrid of ResNet-50 and ViT. The evaluation process includes the use of accuracy metrics, confusion matrices, and learning curves in order to assess the accuracy, generalization potential, and explainability of each model.

The hybrid ResNet-50 and Vision Transformer model comes out at the top with an accuracy of 99.50%, the highest in the category of distinguishing normal situations from stone present ones. As shown in Fig. 6, the confusion matrix of the hybrid model demonstrates its reliability in diagnosing kidney stones with 1,772 true negatives and 1,036 true positives, the model accurately identifies both normal and stone cases. The hybrid model yields low rates of both false positive and false negative diagnosis signifying reliability of the diagnosis which are 2 and 12 respectively. This high accuracy is attributed to the architecture of the model as it combines convolution layers of Resnet-50 for feature extraction and utilizes self attention mechanism of the vision transformer to capture both local and global features. The learning curve of this model as illustrated in

Fig. 7 displays a smooth convergence with very low overfitting, this shows that the model is capable of learning the relevant features in the data presented to it instead of learning the training data by rote hence improving its applicability in new cases.

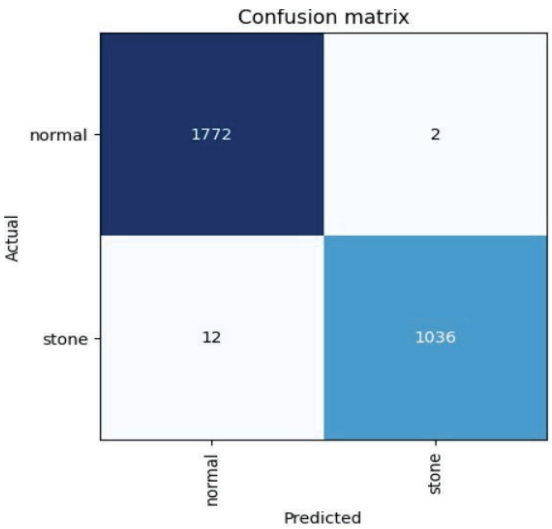

Fig. 6. Confusion Matrix

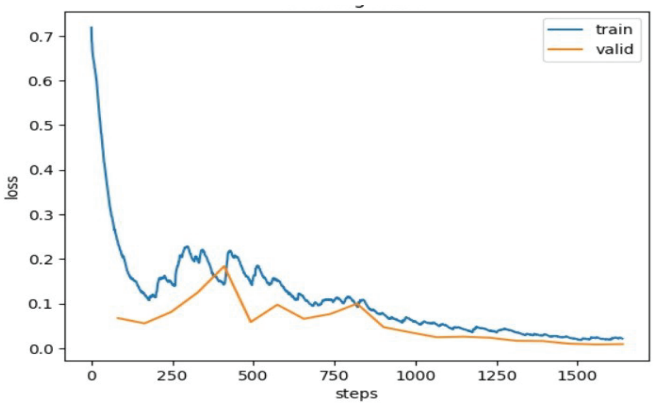

Fig. 7. Learning Curve: Training and Validation Loss Over Steps

TABLE II. COMPARATIVE STUDY OF DIFFERENT MODELS

SL No	Model	Accuracy (%)	Loss Function (%)
1	Resnet+Vit	99.50	2.83
2	XResnet	99.36	60.26
3	VGG16	91.0	5.83
4	AlexNet	97.70	19.04
5	MobileNet	98.69	7.44
6	Swin Transformers	98.13	5.63

According to Table II comparative study the XResNet achieves 99.36% accuracy coming in second to the hybrid model. Its deep structure allows it to pick up on small details in CT scans, which makes it very helpful for medical imaging. But because it doesn't have the Vision Transformer's self-attention feature, it might not catch bigger context clues as well leading to lower accuracy than the hybrid model.

AlexNet reaches 97.70% accuracy, which is pretty good but not as high as the newer models. Its simpler design limits how well it can spot complex patterns in kidney stones. Looking at AlexNet's learning curve, we see signs of overfitting. The gap between training and validation loss starts to grow at some point suggesting it has a hard time applying what it's learned to new data.

MobileNet hit 98.69% accuracy showing it can grasp general patterns but struggles with complex medical images. Its lightweight design puts speed over depth, which might lead to less accurate diagnoses. The learning graph points to overfitting, as test results start to differ from training results later on hurting its ability to work well on new data.

The Swin Transformer scored 98.13% accuracy striking a nice balance between being precise and working well on different data. It's better at understanding how things relate in CT scans than simpler models, thanks to its self-attention tricks. Still, it's not as accurate as the mixed model.

VGG16 scored the lowest accuracy at 91.0%. This lower accuracy stems from its basic and shallow design, which doesn't have the complex workings found in ResNet, Vision Transformers, or Swin Transformers. As a result, VGG16 struggles to handle the complex feature hierarchies needed to spot kidney stones.

To assess the comprehensibility of the hybrid ResNet-50 and Vision Transformer model, we used Grad-CAM visualizations. Fig. 8(a) shows the original CT scan image and Fig. 8(b) shows the Grad-cam overlay on the CT scan image. These visualizations proved that the model could spot kidney stones and pinpoint their location in CT scans. This ability to

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locate stones makes the hybrid model useful in hospitals. It helps doctors see where to look, which leads to more focused diagnoses and treatments.

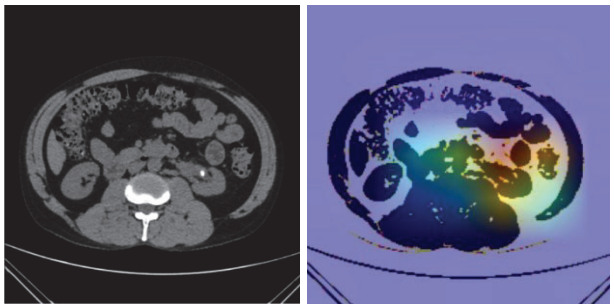

Fig. 8. (a): Original Image (b): Gradcam visualization

VI. Conclusion

In conclusion, this study shows how effectively the hybrid ResNet-50 and Vision Transformer model is in identifying kidney or urinary stones from CT scan images with high diagnostic precision. By combining the strengths of convolutional and self-attention mechanisms, this model offers a non-invasive reliable way to differentiate between normal and stone cases with few false positives and negatives. The hybrid model outperforms traditional models when handling complex image patterns and imbalanced data. It also provides interpretable results through Grad-CAM visuals making it suitable for clinical use.

This research impacts on more than just accurate diagnosis to enhance patient care. It can be used to help plan treatments, guide decisions about interventions, and sort cases by priority. In the end, the creation of such a sophisticated diagnostic tool improves patient outcomes in kidney stone management and streamlines healthcare delivery, particularly by boosting diagnostic confidence and facilitating focused clinical interventions.

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